

Assignment 1: Applying CNN to Your Own Image Dataset

Reference

Follow the TensorFlow CNN tutorial:

[TensorFlow – Convolutional Neural Network \(CNN\)](#)

The purpose of this assignment is to apply the CNN concepts from the tutorial to your own image classification problem.

Task 1 – Application and Dataset

Choose an image classification application and your own dataset.

Briefly write:

1. Application: What are you trying to classify?
2. Dataset: What dataset are you using?
3. Classes: What are the categories/classes?
4. Number of images: How many images are available?
5. Show a screenshot of your original dataset with some example images.

Task 2 – Data Preprocessing

Apply appropriate preprocessing to your dataset before training the CNN.

For example:

- Resize images
 - Normalize pixel values
 - Convert labels
 - Split data into training/testing sets
 - Apply data augmentation, if needed
1. Write what preprocessing steps you applied and why.
 2. Show some images before and after preprocessing.
 3. Also include a screenshot of your preprocessing code/output.

Task 3 – Apply the TensorFlow CNN

Use the CNN approach from the provided TensorFlow tutorial and adapt it to your dataset.

You may need to change:

- Input image size
- Number of classes
- CNN layers
- Number of filters
- Training parameters

Do not simply run the original CIFAR-10 example.

1. Generate and include: `model.summary()`
2. Briefly explain your CNN architecture.

Task 4 – Optimization Technique

Select an optimizer for training your CNN.

Examples include:

- SGD
- Adam
- RMSprop

Answer the following:

1. Which optimizer did you use?
2. Why did you choose this optimizer?
3. What learning rate did you use?
4. How does the optimizer update the model parameters during training?
5. How can you determine whether the optimizer has converged?

Optimization Convergence

Use your training results to discuss the convergence of the optimizer. Include your training and validation loss curves and explain:

1. Does the loss decrease as training progresses?
2. Does the loss eventually become stable?
3. Does the validation loss continue decreasing or start increasing?
4. At approximately which epoch does the model appear to converge?
5. Is there any indication of unstable training or overfitting?

Your discussion should be based on your actual training curves.

Task 5 – Training Results

Generate:

1. **Training vs. validation accuracy graph**
2. **Training vs. validation loss graph**

Briefly discuss:

1. Did the model learn successfully?
2. Did the optimizer converge?
3. Is there any sign of overfitting or underfitting?
4. How well did the CNN perform on your dataset?

Task 6 – Effect of CNN on Your Dataset

Discuss the results in a short paragraph. Explain:

1. What patterns/features did the CNN learn?
2. Which classes were classified well?
3. Which classes were difficult?
4. What could be improved?

Task 7 – Future Extension: Transfer Learning

Choose ONE CNN architecture that you would consider using in the future for transfer learning.

Choose one:

AlexNet, VGG16, ResNet, Inception, MobileNet, EfficientNet, etc.

Briefly answer:

1. Which architecture did you choose?
2. Why did you choose it for your dataset?
3. How could transfer learning with this model potentially improve your results?

You do not need to implement the transfer-learning model. This is only a future-work direction.

Final Submission

Your report should contain:

1. **Application and dataset**
2. **Original dataset screenshot**
3. **Preprocessing steps + before/after images**
4. **CNN implementation**
5. **Model summary**
6. **Optimizer + reason for selection**
7. **Optimization/convergence discussion**
8. **Accuracy and loss graphs**
9. **CNN results/discussion**
10. **Future transfer-learning architecture**

Important

You must use **your own dataset** and explain the changes you made to the TensorFlow tutorial for your application.